Hot Wings: The Impact of Large Appendages on the Heat Loss Potential and Migratory Decision-Making of Humpback Whales

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Abstract

Marine mammal thermoregulation depends primarily on heat flux through the well-insulated body wall and variable blood-flow through minimally insulated appendages (e.g.,

flippers, fins, flukes). Models of heat loss have been developed for many species, but the majority are focused on the body. Humpback whales (*Megaptera novaeangliae*) have large appendages relative to their body size, positioning them as a unique species to study the impact of these "thermal windows" on overall heat loss potential and resilience to rising oceanic temperatures. By combining heat loss potential from the body and the appendages, we were able to estimate the thermoneutral zone and sustainable metabolic range of humpback whales spanning a range of body sizes in both warm (Hawai'i) and cold water (Southeast Alaska) environments. Heat loss was higher in Hawai'i with thermal windows "open" (5803 ± 1432 MJ day^{-1}) than in Southeast Alaska with thermal windows "closed" (3158 ± 994 MJ day^{-1}), representing the upper and lower bounds of heat dissipation and retention, respectively. The appendages accounted for 79.8 ± 2.2 % of total conductive heat loss with thermal windows "open", helping all but the largest individuals in Hawai'i to avoid hyperthermia. Small calves, meanwhile, required metabolic rates above basal in Southeast Alaska, even with thermal windows fully "closed". Interpreting this elevation as a distinct energetic cost resulted in an additional heat loss burden of 6938 MJ. These results demonstrate that the large appendages of humpback whales provide a powerful mechanism for regulating heat loss, allowing individuals to avoid both hyperthermia and hypothermia across contrasting thermal environments.

Introduction

Maintaining a favorable balance between energetic gain and expenditure is critical for the continuation of life. For animals, energetic intake comes from a singular source (foraging), but energetic costs are often varied and include complex behavioral and physiological processes such as locomotion (for motile species) (Alexander, 2005; Schmidt-Nielsen, 1972), prey capture (Bolnick and Ferry-Graham, 2002; Griffiths, 1980), digestion (Armstrong and Blaxter, 1956), somatic growth (Christiansen et al., 2022; Millward et al., 1976; van Aswegen et al., 2025c), reproduction (Bell, 1980; Remick, 1992), thermoregulation (Scholander et al., 1950), and basal metabolism (He et al., 2023; Kleiber, 1975; Spaargaren, 1994). For species that can be housed under human care, these costs can oftentimes be quantified holistically or separated out into their component processes (Allen et al., 2022; Fahlman et al., 2025; Lindström et al., 1999; Williams and Davis, 2024). For larger or cryptic species that cannot be housed under human care, or for situations where species cannot be reliably observed without significant changes to behavior, energetic estimates are often modeled through a patchwork of quantifiable parameters and up-to-date approximations from more tractable species or historical records (Boyles and Brack, 2009; Gough et al., 2022; Pirotta et al., 2018).

The infraorder *Cetacea* includes many large-bodied, highly mobile species that cannot be ethically or logistically maintained in human care. Field experiments on large, free-swimming cetaceans require ideal conditions, large swaths of open ocean, and cutting-edge technology (e.g., biologging tags, unmanned aerial systems [UAS] or drones, acoustic monitors) to interpret changes in behavior (Castellote et al., 2012; Curé et al., 2013; Curé et al., 2015; Southall et al., 2016). These constraints have made large cetaceans strong

candidates for computational modeling, both to generate energetic baselines (Christiansen et al., 2023; Fahlman et al., 2025; Goldbogen et al., 2019a; Gough et al., 2022; Gough et al., 2025) and to explore the potential physiological consequences and energetic costs associated with disturbance (Christiansen et al., 2025; Czapanskiy et al., 2021; Hin et al., 2019; Pirotta et al., 2018).

Long-distance migrations between cold foraging areas and warm breeding habitats add significant complexity to energetic modeling of large whales. Understanding the annual energetic cycle of these species requires empirical data or accurate estimations of various costs for both sides of the migratory route. Given the substantial differences in water temperature between the two locations (as much as ~30°C), thermoregulation is likely among the most energetically costly processes affected by migration, requiring a suite of adaptations to either conserve or dissipate heat in response to environmental conditions (Favilla et al., 2024). Researchers have posited multiple behavioral reasons for the migratory life history of large whales, including predator avoidance in the early stages of life, enhanced parasite removal, seasonality of resources, or the possibility of a more favorable thermal environment during early development (Clapham, 2001; Corkeron and Connor, 1999; Pitman et al., 2020). This latter hypothesis treats thermoregulation as a distinct energetic cost. However, recent studies (Christiansen et al., 2026; Sumich, 2021) have challenged this interpretation, suggesting instead that heat loss is simply a byproduct of metabolism. These studies have found the heat loss of newborn calves to be lower than their estimated FMR, implying that thermoregulation is unlikely to drive migration provided the calves can adequately dissipate excess metabolic heat (Christiansen et al., 2026; Sumich, 2021).

The majority of cetacean heat loss is thought to be modulated through regional heterothermy in the extremities (i.e., flippers, tail flukes, and dorsal fin), with a counter-current heat exchange system that either restricts or facilitates the movement of warm blood near to the skin's surface (Scholander and Schevill, 1955). This system consists of rosettes of veins surrounding a central artery that permits heat exchange to route warm blood back to the body. When a large heat load must be dissipated, a rise in blood pressure increases the diameter of the central artery, collapsing the surrounding veins and distributing blood to superficial veins beneath the skin. Body and appendage sizes should, therefore, have a significant impact on thermoregulation. Bergmann's rule suggests that larger animals with lower surface-to-volume ratios will inhabit colder environments at higher latitudes, while Allen's rule suggests that species living in colder environments will have smaller appendages to decrease excess heat loss (Ray, 1960; van Aswegen et al., 2019). Among the large whales, the humpback whale (*Megaptera novaeangliae*) exists at an intersection of these two rules, having large body size (~30,000 kg; ~12-14 m in length; van Aswegen et al., 2025c), but also disproportionately large flippers (~30% of body length) (Fish and Battle, 1995; Kahane-Rapport and Goldbogen, 2018). It is also one of the few large whale species that migrates all the way from near-freezing polar waters to warm tropical waters and back, thereby experiencing an exceptionally wide range of thermal environments.

Models of thermoregulation in large whales have traditionally focused on body heat loss, with thermal windows treated as areas of minimal heat loss that remain "closed" (Christiansen et al., 2026; Glarou et al., 2025a; Sumich, 2021). While this approach captures the capacity to conserve heat, it does not account for the potential contribution of thermal windows to dissipating excess metabolic heat. This may be particularly important for humpback whales, whose disproportionately large flippers and flukes provide substantial surface area for heat exchange and may expand the range of metabolic activity that can be thermally sustained.

Building on the large body of physiological and anatomical work performed on small cetacean species in human care, as well as the models generated for a variety of larger cetacean species, we sought to combine uniquely broad UAS-derived morphometric and biologging tag-derived kinematic datasets from humpback whales across a range of body sizes in both Hawai'i and Southeast Alaska into a holistic computational model of heat loss potential under two conditions: with thermal windows "open" and "closed". From this, we hoped to provide a novel estimate of metabolic scope and the thermoneutral zone. We hypothesized that the large appendages of humpback whales would substantially increase heat dissipation when thermal windows were open, particularly benefiting large individuals in warm water. Conversely, we predicted that small-bodied calves would face greater thermal constraints in cold water, requiring higher minimum metabolic rates to maintain thermal balance. Together, these comparisons allowed us to assess how body size and environmental temperature shape thermoregulatory constraints across the humpback whale migratory range.

Methods

Morphometric Variables From UAS-Photogrammetry

For the majority of our analyses, we used a dataset of morphometrics (n = 5929 individuals) collected by UAS-photogrammetry on humpback whales between 2018 and 2025 across Hawaiian (breeding ground; n = 3701) and Southeast Alaskan (foraging ground; n = 2228) waters. Additional details on the UAS protocols and data collection methodology pertaining to this dataset have been provided by van Aswegen et al. (2025b; 2025a; 2025c).

Morphometric measurements for each animal included the total body length from the anterior point of the rostrum to the tail notch (TL; in m) and 19 distinct width measurements taken at 5% intervals along the length of the animal. From the combination of these length and width measurements, we estimated the body volume (BV; in m^3) following the approach of Christiansen et al. (2019), applying body height-width ratios measured in van Aswegen et al. (2025a). The body condition of each animal was estimated using residuals calculated from the log-log linear relationship between BV and TL, fitted separately for adult female, male, and unknown-sex whales (van Aswegen et al. 2025c). We estimated total body mass (M_{body}; in kg) following Christiansen et al. (2022), by

partitioning the body volume of an animal into tissue types that scale with TL - muscle, bone, viscera - and blubber volume, which varies with body condition. Body surface area (S_{body}; in m^2) was calculated using the approach of Glarou et al. (2025a), using our estimates of TL, body width, and body height. Calf age was estimated following van Aswegen et al (2025c), by modeling birth size and postnatal growth rates as functions of maternal TL. Approximate birth dates were inferred from the predicted BV at birth and daily growth rates derived from repeated measurements or from the relationship between maternal TL and calf BV growth rates. Figure S1 provides a visualization for a subset of relevant morphometric measurements.

We used a subset of UAS images to determine the surface area of the pectoral flippers (S_{flippers}; in m^2) and tail flukes (S_{flukes}; in m^2). In all images, animals were at the surface with their flippers and flukes extended and flat. We used ImageJ v1.54g (Schindelin et al., 2012) to scale each image to the known TL of the animal, then outline each extremity to determine the dorsal area ($S_{flippers}$ and S_{flukes}; in m^2). The left and right sides of each pair of extremities were measured separately, then averaged together to account for measurement error. To determine the overall area of each pair of extremities, the "single side" values for $S_{flippers}$ and S_{flukes} were quadrupled. For each individual in our subset of UAS images, we regressed relevant morphometric values ($S_{flippers}$ and S_{flukes}) against TL (Figure S2) and used those predictive models to estimate the extremity areas for each individual in our broader dataset.

Body and Appendage Blubber Thickness

We estimated the body blubber thickness (BT_{body}, m) as a same-thickness sheet surrounding the entire body (without appendages) using a regression equation for humpback whales taken from Glarou et al. (2025a):

$$BT_{\text{body}} = TL * \left(0.0119345 - (0.0003168 * TL)\right) \quad (1)$$

To determine the blubber thickness of the appendages ($BT_{\text{appendage}}$), we obtained flippers and/or flukes from stranded individuals representing multiple species, including the pygmy killer whale (*Feresa attenuata*), striped dolphin (*Stenella coeruleoalba*), harbor porpoise (*Phocoena phocoena*), and humpback whale (measurements are outlined in Table S1). We divided each appendage at three cross-sectional positions (approximately 25%, 50%, and 75% along the span) and measured the average blubber thickness from 12 positions around the periphery of the chord at each position. To determine a single measure of blubber thickness for each appendage, we averaged our measurements (36 in total) for that appendage. For the harbor porpoise and humpback whales in our dataset, we only had single mid-span cross-sectional images. For individuals where we had a flipper and fluke, blubber thicknesses were found to be similar between the two. As a result, we incorporated measures from flippers and flukes into a single "appendage" measurement. For each individual, we also had an estimate of TL, allowing us to create a regression between $BT_{\text{appendage}}$ and TL and use that to determine the $BT_{\text{appendage}}$ for each individual in our broader morphometric dataset (Figure S3).

Conductive Heat Loss

We used the "flat plate" model first introduced by Blix and Folkow (1992) to determine the heat loss (HL_{body}; in W) from the body of an animal to its environment using the equation:

$$HL_{\text{body}} = \frac{k * S_{\text{body}} * \left(T_{\text{body}} - T_{\text{ambient}}\right)}{BT_{\text{body}}} \quad (2)$$

where k was the conductivity of blubber (W m^{-1} °C^{-1}), T_{body} and $T_{ambient}$ were the internal body temperature of the animal and the ambient temperature of the water, respectively (°C). Our model assumed that there was sufficient forced convection to create a negligible boundary layer and maintain the full temperature gradient between T_{body} and $T_{ambient}$.

To determine the heat loss potential from the flippers (HL_{flippers}; in W) and flukes (HL_{flukes}; in W) of the animal, we used the same heat loss equation (Equation 1) and substituted in values specific to each extremity:

$$HL_{\text{flippers}} = \frac{k * S_{\text{flippers}} * \left(T_{\text{flippers}} - T_{\text{ambient}}\right)}{BT_{\text{appendage}}} \quad (3)$$

$$HL_{\text{flukes}} = \frac{k * S_{\text{flukes}} * \left(T_{\text{flukes}} - T_{\text{ambient}}\right)}{BT_{\text{appendage}}} \quad (4)$$

Ventilatory Heat Loss

We determined the average ventilation rates r_v (breaths min^{-1}) of humpback mothers (n=20) and calves (n=32) over a period of hours using methods outlined by Blawas et al. (2025) and accelerometry tag data (Customized Animal Tracking Solutions) collected between 2020-2025 in Hawaiian waters from animals in primarily resting and transiting behavioral states (Hollers et al., 2026). We did not explicitly incorporate variability due to behavioral state for each individual, instead aggregating all breaths throughout a tag deployment into a single averaged mean. Figure S4 shows the relation between r_v and TL. Using this regression, we estimated r_v for each of the individuals in our morphometric dataset, then used those r_v values to estimate the heat lost to warm expired air during ventilation using an equation from Sumich (2021):

$$HL_{\text{vent}} = \frac{r_v}{60} * \left(143.5 - 1.37 * T_{\text{ambient}} - 0.07 * T_{\text{ambient}}^2\right) \quad (5)$$

Overall Heat Loss Potential

For our overall determination of heat loss potential in each environment, we considered two scenarios: one where the thermal windows (i.e., the flippers and flukes) were "open" and actively contributing to heat loss, and one where those same thermal windows were fully "closed", allowing the animal to conserve heat (Figure 1B-C). In the "open" condition, the temperature of the flippers and flukes ($T_{flippers}$ and T_{flukes}) was set equal to T_{body},

while in the "closed" condition, the temperature of the extremities was set equal to the $T_{ambient}$, resulting in a heat loss potential of zero (George et al., 2021). In either scenario, our overall heat loss was determined by combining conductive and ventilatory heat loss using the equation:

$$HL = HL_{\text{body}} + HL_{\text{flippers}} + HL_{\text{flukes}} + HL_{\text{vent}} \quad (6)$$

We did not have the data to estimate intermediate heat loss scenarios, and instead had to rely on our binary extremes (i.e., fully "open" and fully "closed). To account for unknown sources of variability in our resultant heat loss estimates, we performed a Monte-Carlo simulation (100 iterations with replacement) for each individual in our dataset. In each iteration, we randomly chose values from defined ranges for a series of variables. We used relevant body length (TL) regressions to anchor range midpoints for variables such as body mass (M_{body}), surface area (body, S_{body}; flipper, S_{flipper}; fluke, S_{fluke}), body blubber thickness (BT_{body}), and ventilation rate (r_v). For M_{body} and S_{body}, we chose ranges extending ±10% from those midpoints based on work by van Aswegen (2025c) and our own dataset (Figure S2A), respectively. For S_{flipper} and S_{fluke}, those same ranges were extended ±25% from each midpoint, once again based on our own dataset (Figure S2B-C). For BT_{body}, we used a range of ±10% based on estimates of humpback blubber thickness taken from Glarou et al. (2025a). For $BT_{\text{appendage}}$, we used a wider range extending ±50% based on our dataset showing a greater spread in the residuals of the fitted model regression (Figures S3). For r_v, we based our range on the residual standard deviation of the fitted model, equating to ~±30% around the allometric prediction (Figure S4). For k, we chose a fixed midpoint value of 0.2 (Glarou et al., 2025a; Parry, 1949) and a range of ±10%. Finally, we chose fixed midpoint values of 36°C, 5°C, and 25°C for T_{body}, $T_{ambient}$ in Southeast Alaska, and $T_{ambient}$ in Hawai'i, respectively (Deser et al., 2010; Hokkanen, 1990; Kagawa-Viviani and Giambelluca, 2020). These temperatures were also sampled from a ±10% range.

Field Metabolic Rate

We inferred the field metabolic rate (FMR) from the mean ventilation rate (r_v) for each individual in our morphometric dataset using the equation:

$$FMR = \frac{1000 * \left(V_t * V_c * LC_{total} * r_v * F_{O_2} * E_{O_2} * C_{O_2}\right)}{60} \quad (7)$$

where r_v was estimated from TL, V_t was the tidal volume (L breath^{-1}), V_c was the vital capacity, LC_{total} was the total lung capacity (L), F_{O_2} was the fraction of oxygen in atmospheric air, E_{O_2} was the oxygen extraction efficiency, and C_{O_2} was the calorific coefficient of oxygen (kJ L^{-1}). This equation has been used in recent studies to estimate metabolic rates in large whales, including the humpback and southern right whale (*Eubalaena australis*) (Christiansen et al., 2023; Christiansen et al., 2025; Videsen et al., 2023), as well as smaller cetaceans such as the short-finned pilot whale (Gough et al., 2025). Given the various sources of uncertainty in this equation, we included these calculations in our Monte-Carlo simulations (similar to Videsen et al., 2023), generating

100 samples (with replacement) for each individual using paramaterized ranges for each variable. These included a range of 0.4-0.8 for V_t (Fahlman et al., 2016; Wahrenbrock et al., 1974), a range of 0.8-0.9 for V_t (Fahlman et al., 2017; Kooyman, 1973), and a range of 0.3-0.4 for E_{O_2} (Videsen et al., 2023). For each individual, LC_{total} was calculated from body mass using an allometric equation ($0.135 * \left(M_{\text{body}}^{.92}\right)$) (Fahlman et al., 2011; Kooyman, 1973), then varied by ±20% to account for certainty surrounding lung function (Isojunno et al., 2018). Our estimate of r_v was perturbed ±10% from the allometric prediction to reflect individual and contextual variability in breathing patterns. Finally, we set F_{O_2} at 0.2095 and C_{O_2} at 20.1 (Folkow and Blix, 1992).

Critical Temperatures

The upper and lower critical temperatures define the limits of a species' thermoneutral zone (i.e., the range of temperatures within which a consistent basal metabolic rate can be maintained). We followed methods outlined by Watts et al. (1993) to calculate critical temperatures (T_{crit}; in °C) as the ambient temperatures at which conductive HL was equal to the basal metabolic heat generated by the body (BMR; Kleiber, 1975):

$$T_{crit} = T_{body} - \frac{BT_{body} * BMR}{k * S_{body}} \quad (8)$$

To estimate the upper and lower critical temperatures, we performed this analysis with thermal windows open and closed, respectively. In both scenarios, this equation only incorporated conductive HL, though our results allowed us to safely assume that HL_{vent} was negligible in relation to the conductive component.

Statistics

All statistical analyses were conducted in R. Individual median estimates from our Monte Carlo simulations were used for statistical analyses and 5th–95th percentiles were used to represent uncertainty. Relationships between TL and conductive HL were evaluated using analysis of covariance (ANCOVA), with environment and its interaction with TL included to test for differences in HL scaling between environments. Similar ANCOVAs were used to compare whole-body HL between the Hawai'i/open and Alaska/closed regional scenarios and to assess relationships between body condition and mass-specific HL within demographic classes.

The combined effects of TL, $T_{ambient}$, and thermal window state ("open" or "closed") on sustainable metabolic limits were evaluated using a linear mixed-effects model with individual identity included as a random intercept. Upper and lower critical temperatures were similarly evaluated using a mixed-effects model containing TL, critical temperature type (upper or lower), and their interaction, with individual identity as a random intercept. Type III *F*-tests with Satterthwaite approximations for denominator degrees of freedom were used for mixed-effects models.

Relationships between calf age and minimum sustainable metabolic rate were evaluated using generalized additive models (GAMs; R package *mgcv*) with environment-specific smooths of calf age and individual identity as a random-effect smooth, fitted using REML. The cumulative difference in minimum sustainable metabolic requirements between Alaska and Hawai‘i during the first 90 days was estimated by integrating the difference between population-level fitted curves using the trapezoidal rule. Statistical significance was evaluated at $\alpha = 0.05$.

Results

Morphometrics

Our morphometric dataset (n = 5929 individuals) included measurements on 1159 calves (n = 1033 in Hawai‘i; n = 126 in Alaska), 1155 lactating mothers (n = 1013 in Hawai‘i; n = 142 in Alaska), 18 juveniles (n = 17 in Hawai‘i; n = 1 in Alaska), 21 yearlings in Hawai‘i, and 3576 adults of unknown sex (n = 1617 in Hawai‘i; n = 1960 in Alaska). Table 1 presents a subset of important morphometric values for each demographic class and geographic location. Our subset of animals with visually flat flippers and flukes (n = 22) were all from Hawai‘i and covered a similar TL range as our larger dataset (3.51-12.69 m). Relationships between TL and selected morphometric estimates (S_{body}; $S_{flippers}$; S_{flukes}; $BT_{appendage}$; r_v) are shown in the supplementary materials (Figures S2-S4). $S_{flippers}$ and S_{flukes} scaled with TL to the power of 1.8, indicating negative allometric scaling. Consequently, smaller individuals had proportionally greater flipper surface area relative to body size. In our dataset, $S_{flippers}$ and S_{flukes} were found to be 14.6±1.1% and 11.7±0.9% of S_{body}, respectively. The combination of all appendages ($S_{flippers}$ and S_{flukes}) was, therefore, found to be 26.3±2.0% of S_{body} and 20.8±1.3% of total surface area (sum of S_{body}, $S_{flippers}$, and S_{flukes}).

Heat Loss

Broken down into the components of HL, we found that daily conductive heat loss potential averaged across all values of TL was highest from the flippers (Hawai‘i: 2595±606 MJ day^{-1}; Alaska: 7313±1699 MJ day^{-1}) and flukes (Hawai‘i: 2088±484 MJ day^{-1}; Alaska: 5885±1355 MJ day^{-1}). The daily conductive heat loss potential of the body was lower (Hawai‘i: 1120±354 MJ day^{-1}; Alaska: 3158±994 MJ day^{-1}), and the ventilatory heat loss potential was exceedingly low (Hawai‘i: 0.068±0.046 MJ day^{-1}; Alaska: 0.139±0.096 MJ day^{-1}) (Figure S5). When the thermal windows were open, the flippers and flukes collectively accounted for 79.8±2.2% of total conductive heat loss potential, compared with 20.2±2.2% attributable to the body.

Conductive HL increased significantly with TL for the body, flippers, and flukes (all $P < 0.001$; Figure 2). Although conductive HL also differed significantly between environments for all three components (all $P < 0.001$), the relationship between TL and conductive HL did not differ between Hawai‘i and Alaska (body: $F_{1,5925} = 0.29$, $P = 0.593$; flippers: $F_{1,5925} =$

1.16, $P = 0.282$; flukes: $F_{1,5925} = 0.17$, $P = 0.677$). Ventilatory HL similarly increased with TL, but remained negligible relative to conductive HL across body sizes (Figure S5).

Heat flux was higher for the appendages (Hawai'i: 442.7 ± 169.6 W m^2; Alaska: 1248.0 ± 477 W m^2) than it was for the body (Hawai'i: 26.1 ± 7.1 W m^2; Alaska: 73.5 ± 19.9 W m^2), across both geographic regions. Calves displayed higher heat flux in both geographic regions (Hawai'i: 39.4 ± 5.3 W m^2; Alaska: 111.0 ± 14.7 W m^2) than either adults of unknown sex (Hawai'i: 23.1 ± 1.5 W m^2; Alaska: 65.0 ± 3.9 W m^2) or lactating females (Hawai'i: 22.1 ± 0.91 W m^2; Alaska: 62.3 ± 2.1 W m^2).

Combining our conductive and ventilatory heat loss components together, we found mean daily heat loss potential values of 5803 ± 1432 MJ day^{-1} and 1120 ± 354 MJ day^{-1} in Hawai'i for the "windows open" and "windows closed" conditions, respectively. In Alaska, we found a value of 16356 ± 4017 MJ day^{-1} for the "windows open" condition and a value of 3158 ± 994 MJ day^{-1} for the "windows closed" condition. Whole-body HL increased significantly with TL and differed between the Hawai'i/open and Alaska/closed regional scenarios (both $P < 0.001$; Figure 3), with the relationship between TL and HL also differing significantly between scenarios (TL × regional scenario: $F_{1,5925} = 1214.81$, $P < 0.001$).

We found that calves displayed the highest average mass-specific heat loss potential (Hawai'i, 1.27 ± 0.44 W kg^{-1}; Alaska, 3.59 ± 1.23 W kg^{-1}), with lactating females (Hawai'i, 0.23 ± 0.04 W kg^{-1}; Alaska, 0.65 ± 0.10 W kg^{-1}), and adults of unknown sex (Hawai'i, 0.27 ± 0.07 W kg^{-1}; Alaska, 0.76 ± 0.20 W kg^{-1}) maintaining similar values (Figure 4). Mass-specific HL decreased significantly with increasing body condition across all demographic classes (all $P < 0.001$), although the relationship differed significantly between regional scenarios for calves ($F_{1,1155} = 7.64$, $P = 0.00581$), adults ($F_{1,3572} = 71.47$, $P < 0.001$), and lactating females ($F_{1,1151} = 15.1$, $P < 0.001$).

Sustainable Metabolic Range

From our analysis of critical temperatures (T_{crit}), the majority of calves from Hawai'i (74%) were estimated to have lower T_{crit} values between 5°C and 25°C, requiring them to maintain metabolic rates above their assumed Kleiber BMR to achieve thermal balance (Figure 5). The lower T_{crit} for adults fell well below freezing, putting them outside the range of temperatures experienced anywhere on the planet and into the "Zone of Irrelevance" described by Watts et al. (1993). Upper T_{crit} values for many larger adults (longer than ~10 m) fell below our modeled ambient temperatures for Hawaiian waters, requiring metabolic rates below the assumed Kleiber BMR estimates to achieve thermal balance. Both upper and lower T_{crit} decreased significantly with increasing TL, but the rate of decline differed between the two thermal limits (TL × T_{crit} type: $F_{1,5927} = 2.2443068^5$, $P < 0.001$), resulting in a widening of the modeled thermal range with increasing body size.

Assuming that heat loss must equal metabolic heat production to achieve thermal balance, we found that the range of "sustainable" metabolic rates that could be maintained indefinitely (i.e., falling between the lowest heat loss potential of the "windows

closed" condition and the close-to-maximal heat loss potential of the "windows open" condition) was much broader in Alaska and much narrower in Hawai'i (Figure 6). Sustainable metabolic limits varied significantly with TL, $T_{ambient}$, and thermal-window state (all $P < 0.001$), with a significant interaction between TL and thermal-window state ($F_{1,1.7781^{4}} = 3.848707^{4}$, $P < 0.001$). In contrast, the $TL \times T_{ambient}$ ($P = 0.426$), $T_{ambient}$ × thermal-window state ($P = 0.576$), and three-way interactions ($P = 0.928$) were not significant. In Alaska, the smallest calves (TL of <4 m) had a minimum sustainable metabolic rate of $1.6\times$ Kleiber and a maximum sustainable metabolic rate of $10\times$ Kleiber. In Hawai'i, these same small calves had a minimum sustainable metabolic rate of $0.53\times$ Kleiber and a maximum sustainable metabolic rate of $3.4\times$ Kleiber. At the larger end of the size scale (TL of >12 m), animals in Alaska had a minimum sustainable metabolic rate of $0.50\times$ Kleiber and a maximum sustainable metabolic rate of $2.3\times$ Kleiber. Those same animals in Hawai'i had a minimum sustainable metabolic rate of $0.17\times$ Kleiber and a maximum sustainable metabolic rate of $0.78\times$ Kleiber.

Figure 7 shows the minimum sustainable metabolic rate for calves in closer detail relative to estimated calf age. Minimum sustainable metabolic rate varied nonlinearly with calf age in both Hawai'i (edf = 4.12, $F = 698.74$, $P < 0.001$) and Alaska (edf = 7.2, $F = 3094.36$, $P < 0.001$). Heat loss potential under the "windows closed" condition was lower in Hawai'i than in Alaska throughout the first 90 days of life. The cumulative difference between the minimum sustainable metabolic rates in Alaska and Hawai'i over this period was 6938 MJ. Averaged across this period, this corresponds to a daily difference of 77.1 MJ day^{-1}. Somatic growth costs for a humpback calf have been estimated by van Aswegen et al. (2025) at multiple points within the same age range (0 days, 390 MJ day^{-1}; 30 days, 659 MJ day^{-1}; 60 days, 871 MJ day^{-1}; 90 days, 728 MJ day^{-1}). The average daily difference in minimum sustainable metabolic rate between environments is equivalent to 19.8% of estimated growth costs at 0 days of age, 11.7% at 30 days of age, 8.85% at 60 days of age, and 10.6% at 90 days of age.

Discussion

Thermoregulation is a complex physiological process, with small changes in core body temperatures leading to a host of adverse effects among endotherms (Benarroch, 2007). Due to the greatly increased thermal conductivity of water in comparison to air (~23x), these physiological constraints are even tighter for fully aquatic species (Favilla and Costa, 2020). Among these, cetaceans are unique in that they have an extreme body size range - from <50 kg for a vaquita (*Phocoena sinus*) (Hohn et al., 1996) up to >150,000 kg for a blue whale (*Balaenoptera musculus*) (Lockyer, 1976) - even among individuals of the same species (Kahane-Rapport and Goldbogen, 2018). In addition, cetaceans live in waters around the globe, with some species migrating from exceptionally cold environments (0-5°C) to warm tropical waters (~25-30°C) and back within a single year (Kettemer et al., 2022; Kshatriya and Blake, 1988; Modest et al., 2021; Zerbini et al., 2006). Previous research on the thermoregulation of large migratory cetaceans is still somewhat limited,

with the majority of studies focusing on single geographic locations (George, 2009; George et al., 2021) or single demographic classes, such as calves (Sumich, 2021) or adults (Glarou et al., 2025a; Parry, 1949). Watts (1993) and Ryg (1993) incorporated species-level models for larger cetaceans, but the fine-scale morphometric data available to parameterize their models was limited. Christiansen et al. (2026) modelled the heat loss of southern right whales across their full range of body lengths, body conditions, and geographic distribution. However, their model was restricted to conductive heat loss from the body only and did not include heat loss from the flippers or flukes. Our study is the first to explicitly incorporate heat loss from the appendages of the whales, across a wide range of body sizes (from newborn calf to adult) using morphometrics from a large number of animals on both the breeding and feeding grounds.

Model Limitations

In our estimation of heat loss potential from the body, we relied on a simplified view of blubber, with uniform thickness (BT_{body}) and thermal conductivity (k). Both of these variables have been shown to vary due to a number of critical factors (e.g., timing in the annual cycle, location on the body, temperature gradient between the body and ambient environment, amount of blood perfused through the tissue, lipid composition) (Favilla et al., 2024; George, 2009; Lockyer, 1987; Meagher et al., 2008; Miller et al., 2011; Sumich, 2021; Víkingsson, 1990). Modeling BT_{body} solely on TL was necessary for our present model, but changes in body condition may alter blubber stores and, consequently, insulative properties. Recent biologging studies may allow us to better characterize this relationship, as tissue density can be estimated from the gliding dynamics of humpback whales and used as an indicator of lipid stores (Aoki et al., 2021; Narazaki et al., 2018). Combining tag-derived tissue density with UAS-derived body condition and direct measurements of blubber thickness could therefore help determine how changes in energy stores translate into changes in insulation. At present, however, tissue density describes whole-body composition rather than blubber thickness directly, and the relationship between these variables remains insufficiently resolved to incorporate condition-dependent BT_{body} into our model.

In comparison to the body, modeling heat loss potential through the extremities is even more difficult. There are many experimentally-derived values in the literature that we can draw from for the blubber anatomy and thermal dynamics of large whale bodies (Glarou et al., 2025a; Parry, 1949). For the extremities, we relied on the same value of k for both the body and the extremities, and we based our estimation of $BT_{\text{appendage}}$ on the cross-sections of flippers and flukes from multiple species and age/sex classes. These values ranged from 0.47-8.3, though the values from the two ~9 m humpback whales in the dataset were remarkably different (~3 mm in the subadult flipper cross-section versus ~8.3 mm in the juvenile fluke). Unfortunately, there are very few literature sources on the BT of cetacean extremities to compare against. The majority of studies that investigate the flippers or flukes focus instead on other anatomical structures (e.g., skin, collagenous fibers, bone, vasculature), noting only a very thin blubber layer (Felts, 2023; Gavazzi et al., 2024; Gough

et al., 2018; Sun et al., 2010). Only George (2009; 2021) gives actual values for the blubber thickness of the flippers (~20 mm) and flukes (~10-15 mm), but these were for the strongly ice-associated bowhead whale (*Balaena mysticetus*) that spends its entire life in cold arctic waters.

Our model assumes the same thermal dynamics in the appendages as the rest of the body, with an internal temperature (set as the same as the core body temperature) and a single-thickness blubber layer that loses heat uniformly to the environment. Heat loss through the cetacean extremities is tied closely to the anatomical layout of their vasculature (Favilla et al., 2024; Williams et al., 1999), suggesting that heat loss in a more detailed model would take into account areas of higher and lower heat loss potential. As a result of this simplification, we are likely overestimating the heat loss through the extremities. Alternately, a change in the k of blubber on the extremities, associated with factors such as lipid content and composition (Kvadsheim et al., 1996), could lead to additional heat loss that we are unable to capture. Finally, when working with such thin BT values as we included in our model, small changes can have a large impact on model outputs. Additional data will be needed to parameterize these critical variables in future modeling exercises.

Our model also assumes that heat loss through the thermal windows can be reduced to zero when "closed", representing an idealized lower bound of whole-body heat loss. In reality, some heat transfer is likely to persist even under maximal vasoconstriction, and changes in blood flow through the appendages would effectively alter their thermal conductivity (k) between "open" and "closed" states. We did not estimate separate values of k for these states because the extent to which perfusion and countercurrent heat exchange alter heat transfer through humpback whale appendages has not been sufficiently quantified. Consequently, our "windows closed" scenario likely underestimates total heat loss, particularly in cold water.

Finally, our model simplifies the effects of forced convection on heat loss potential, assuming that there is always enough water movement past the skin to nullify the boundary layer and maintain the full predicted temperature gradient between T_{body} and $T_{ambient}$. In reality, warming of the boundary layer is likely to reduce this temperature gradient and, subsequently, the heat loss potential of an animal. For adult humpbacks in Hawai'i that remain stationary for large portions of their day, this could bring them closer to overheating, further necessitating a reduced metabolic rate or additional heat loss mechanisms (e.g., engulfing cool water to engage the highly vascularized tongue and palate (Ford et al., 2013; Heyning and Mead, 1997) that we are unable to capture with our model.

Impact of Thermal Windows on Conductive Heat Loss

Recent models of heat-loss potential in large cetaceans have excluded the extremities (Christiansen et al., 2026; Glarou et al., 2025a; Sumich, 2021), focusing instead on core body heat loss and assuming that, provided FMR exceeds body heat-loss potential,

animals can maintain regional heterothermy or modulate thermal windows as needed to maintain body temperature within a thermoneutral range (Sumich, 2021). We sought to test this assumption by developing a model that includes a simple estimation of heat loss potential through the extremities. With this holistic model, we could estimate the previously defined lower bound of heat loss potential (with the thermal windows "closed") as well as the presumed upper bound of heat loss potential (with the thermal windows fully "open"), the latter of which may be defined as the maximum thermoregulatory cost that the animal could sustain without overheating (Hokkanen, 1990).

Previous research by Meagher et al. (2008) described the heat flux through different areas of the body in both summer and winter for bottlenose dolphins (*Tursiops truncatus*) in Florida waters. They found only slightly higher summertime values for the flippers (155.3 ± 15.1 W m^2) and flukes (126.2 ± 12.4 W m^2) relative to the body (109.1 ± 5.3 W m^2). Our equivalent heat flux values for Hawai'i showed a much greater discrepancy between the flippers (442.7 ± 169.6 W m^2) and flukes (442.7 ± 169.6 W m^2) and the rest of the body (26.1 ± 7.1), with heat loss through the extremities accounting for $79.8 \pm 0.52\%$ of total conductive heat loss potential in Hawai'i. This percentage was far greater than the 10-15% predicted by Ryg et al. (1993). Their prediction was based primarily on species with much smaller appendages, relative to body size. For a large cetacean, they predicted the surface area of appendages to be ~10% of total surface area, while our humpbacks displayed appendages that were $26.3 \pm 2.0\%$ of body surface area and $20.8 \pm 1.3\%$ of total surface area.

The uniquely large appendages of humpback whales result in enhanced maneuverability and allow for a wide repertoire of foraging behaviors (e.g., bubble-netting) on the feeding grounds (Fish and Lauder, 2006; Nemeth et al., 2025; Szabo et al., 2024; Woodward et al., 2006), but they likely also convey a higher heat loss potential on the warm breeding grounds, resulting in a greater scope for movement or energetically intense behaviors such as competitive male "heat runs" (Tyack and Whitehead, 1982) or repeated breaches (Segre et al., 2020). Bergmann's rule (that larger animals should inhabit colder waters) and Allen's rule (that animals residing in colder climates have smaller appendages) have been suggested as reasons for the appendage-to-body size ratios in various cetacean species (Adamczak et al., 2020; Favilla et al., 2024; Torres-Romero et al., 2016), but humpbacks appear to defy these rules by migrating between both tropical and arctic climates while having the largest appendages (relative to body size) of any cetacean (Kahane-Rapport and Goldbogen, 2018; Woodward et al., 2006). The large size of humpback whale appendages may therefore be key to their ability to tolerate wide variation in water temperatures without risking either hyperthermia or hypothermia.

The large body sizes of humpback whales, and their subsequently low surface-to-volume ratio, limit their capacity to dissipate excess heat across the body in tropical climates, increasing the potential importance of appendages as thermal windows. If our model predicted similarly low heat flux values for both the body and the extremities, adults would be incapable of dispersing enough heat to keep from overheating in Hawaiian waters. Heat flux estimates from the fore- and hind-flippers of harp seals showed higher values (~100-

300 W m^2) at ambient temperatures of 23-24°C, with short-duration increases (up to 20-fold) measured for one or both hind-flippers (Kvadsheim and Folkow, 1997). If the countercurrent heat exchange system in cetaceans works similarly, it suggests that our higher extremity heat flux values are not outside the realm of possibility. It is also possible that whales can increase heat dissipation from the main body by elevating blood flow to the outer blubber layer, thereby increasing k (Kanwisher and Sundnes, 1966). In this scenario, whales would rely less on their appendages for heat dissipation in warmer waters or during periods of elevated activity.

Finally, the relative importance of the thermal windows may also change across body size as individuals grow. Both flipper and fluke surface area exhibited negative allometric scaling relative to TL, with scaling exponents of 1.84 (95% CI = 1.71–1.97) and 1.83 (95% CI = 1.71–1.95), respectively. Calves and juveniles therefore have proportionally larger appendages than adults, providing greater relative capacity for heat dissipation when their thermal windows are open. However, this additional capacity may provide little thermoregulatory benefit to smaller individuals in warm water, where they are less likely than larger adults to approach their maximal heat loss threshold. Instead, relatively large appendages may represent a greater thermal liability in cold water, where residual heat loss through these regions could disproportionately increase heat loss in smaller individuals.

Effects of Body Size on Metabolic Rate

Estimating metabolic rates in large cetaceans has remained a persistent challenge, but our understanding of this important physiological metric underlies our broader knowledge of species health and resilience. For smaller species that can be housed under human care, researchers have estimated basal metabolic rates ranging from ~1.5-3.5 times the Kleiber (1975) “mouse-to-elephant” allometric curve (He et al., 2023; Noren and Rosen, 2023; Williams, 2022) - much higher than similarly-sized terrestrial counterparts. For larger-bodied species such as the humpback whale, basal metabolic rates cannot be directly measured through any of the typical experimental pathways (e.g., respirometry, monitoring of heart rate, ingestion of doubly-labeled water). Instead, we must rely on anatomically-derived models (Lockyer, 1981) and inform them with any physiological data that we can obtain (Goldbogen et al., 2019b).

These models have, so far, produced estimates broadly consistent with those for smaller cetaceans, suggesting a potential allometric relationship and possibly elevated BMR in larger species (Williams, 2022). However, this remains a matter of considerable debate: while some studies report higher BMR in aquatic mammals relative to terrestrial species, others find no such elevation once methodological and analytical issues are accounted for (e.g., Glarou et al., 2025b; He et al., 2023; Lavigne et al., 1986; McNab, 1988; White, 2010). Thus, whether large cetaceans truly exhibit elevated BMR remains unresolved. However, for the purposes of this study, we assume that BMR scales allometrically as predicted by Kleiber.

If we assume that the entire metabolic budget will eventually need to be offloaded as heat, we can use our model of heat loss with the thermal windows "closed" and "open" as proxies for the minimum and maximum sustainable metabolic rates, respectively. FMR values for smaller cetaceans have been estimated at ~3.5 times the Kleiber prediction (Rojano-Doñate et al., 2018), and the FMR of calves and juveniles is thought to be higher than for adults (Innes et al., 1987; Watts et al., 1993). For the animals on the feeding grounds in Alaska, the metabolic "scope" between minimum and maximum heat loss potential ranged from 1.6 to 10 times the Kleiber prediction for the smallest animals (<4 m) and from 0.50 to 2.3 times the Kleiber prediction for the largest animals (>12 m). These values captured the full range of literature predictions at the lower end of the body size scale, suggesting that calves and juveniles could sustain even higher metabolic rates, if necessary. For larger animals, heat loss potential in the "open" condition is below the literature-based FMR prediction of ~3.5. As stated previously, small changes in the parameters of our appendage-based heat loss model could account for these discrepancies. Individuals may also be relying on additional avenues of heat loss. Research on gray whales (*Eschrichtius robustus*) has suggested that the intake of water into the mouth - with or without prey - may serve a thermoregulatory function, with the highly vascularized upper palate and tongue acting as thermal windows (Ford et al., 2013; Heyning and Mead, 1997). The implications of this behavior are difficult to incorporate into a heat loss model, but any additional heat loss would serve to increase the maximal sustainable metabolic rate, allowing for more exertion with less chance of overheating. In comparison to Alaska, we found a much smaller range of sustainable metabolic rates for adults in Hawai'i (from 0.17 to 0.78 times the Kleiber prediction), but these animals are fasting and spending a large portion of their time resting (Bejder et al., 2019; Christiansen et al., 2025), both factors that curtail expenditure and could maintain FMR values close to - or below - basal levels estimated from Kleiber.

Migratory Decision-Making

Corkeron and Conner (1999) summarized multiple hypotheses for the long-distance migrations observed in large cetacean species such as the humpback whale, concluding that predator avoidance (i.e., leaving waters that contain large numbers of killer whales [*Orcinus orca*] capable of preying on calves) was the likeliest reasoning. Clapham (2001) challenged this hypothesis, noting the absence of documented killer whale attacks in major humpback feeding and breeding areas. He acknowledged, however, that predation may occur along migratory routes (Mehta et al., 2007) and that extended residency on the breeding grounds may allow calves to reach sizes that reduce vulnerability to otherwise fatal attacks. The thermoregulatory explanation - that calves require warmer waters on the tropical breeding grounds to maintain thermoneutrality and maximize energetic investment in somatic growth - was discounted by Corkeron and Conner (1999), citing multiple models of calf thermoregulation that have been performed through the years (Kanwisher and Sundnes, 1966; Watts et al., 1993). In their review and in the subsequent Clapham (2001) commentary, it was noted that additional detail on the anatomical

structure and thermal dynamics of cetacean blubber would be necessary to find a sufficient and lasting answer to the migration question.

In the years since, many of these variables have been fleshed out for a number of species (see Glarou et al. (2025a) for a comprehensive overview), and additional models of heat loss have been tested (Christiansen et al., 2026; Sumich, 2021), with the majority agreeing with the assertion by Corkeron and Conner (1999) that heat loss was likely not a major factor in migratory decision-making. The study by Sumich (2021), for example, compared heat loss of gray whale calves against estimates of resting metabolic rate and found that, even under conditions of extreme lipid-depletion, metabolic rates were sufficient to cover heat loss. Similarly, Christiansen et al. (2026) showed that the large body size, and consequent low surface-area-to-volume ratio, of southern right whales, even at birth, is sufficient to offset heat loss, such that only a thin layer of blubber is required to keep thermal loss below their FMR. In our dataset, a small calf would fall well outside its thermoneutral zone if its metabolism were near the basal Kleiber (1975) prediction; however, this scenario is unlikely, as FMR values for small calves (<4 m) were estimated to be 4.3 times the Kleiber prediction. Notably, the minimum sustainable metabolic rate for a calf born in Alaska's colder waters was substantially higher than for a comparably sized calf born in Hawai'i, differing by 6938 MJ. If heat loss is considered to be a necessary physiological side-effect of metabolic heat production (as described by Sumich et al. (2021) and others), then that increase in heat loss would be insignificant so long as it remains below the amount of metabolic heat generated by the animal. But if thermoregulation is thought of as a distinct and realized cost to the animal, then increasing the heat losses in Alaska would necessitate either cuts to the energetic budget in other areas (e.g. somatic growth), as has been posited by Clapham (2001), or increases in the overall metabolic rate to offset increased losses.

Interestingly, one of the few large cetacean species that remains in cold waters throughout the year, the bowhead whale (*Balaena mysticetus*), has been shown to have similar calf growth rates to the humpback - both are born at ~4 m in length, and grow to ~8 m within the first year (George, 2009; van Aswegen et al., 2025c) - suggesting that a non-migratory life history strategy does not necessarily result in slower growth during early life. Bowhead whales are also suspected to have lower core body temperatures (~34°C), resulting in a smaller temperature gradient and subsequently reduced heat loss from the animal to its environment. The lack of large appendages, however, likely limits the bowhead from maintaining a more migratory lifestyle and surviving in warmer environments.

Conclusions

Thermoregulation is a critical physiological function for all marine endotherms, but especially so for humpback whales and other large cetacean species that migrate between cold arctic feeding grounds and warm tropical breeding grounds throughout the annual cycle. Our results demonstrate that the disproportionately large appendages of humpback whales substantially expand their heat-dissipation capacity on warm breeding grounds, leading to a decreased risk of overheating at large body sizes. For calves, our results show

603 that colder environments impose a higher minimum sustainable metabolic rate and
604 increase the total amount of heat loss during early development (6938 MJ), although these
605 costs remains well below estimated FMR values for similarly-sized animals. Whether this
606 excess heat loss constitutes a meaningful energetic burden or simply a byproduct of
607 metabolic heat production remains unanswered. Regardless, our findings highlight
608 thermoregulation as an important axis of energetic scope that interacts with body size,
609 condition, life stage, and habitat. Incorporating full-body heat loss potential - including the
610 contribution of dynamic thermal windows - into bioenergetic models is essential for
611 understanding migratory decision-making, developmental constraints, and the
612 vulnerability of large whales to future ocean warming.

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Competing interests

The authors declare no competing or financial interests.

Author contributions

Conceptualization: W.T.G., L.B.; Data Curation: W.T.G., M.v.A., L.E., A.S., J.J.C., A.H., F.E.F., K.L.W., J.P., N.H.; Formal Analysis: W.T.G., M.v.A., F.E.F., J.P., N.H.; Funding Acquisition: A.S., J.J.C., K.L.W., L.B.; Investigation: W.T.G., M.v.A., L.E., A.S., J.J.C., A.H., F.E.F., K.L.W., J.P., N.H., L.B.; Methodology: W.T.G., M.v.A., A.F., F.C., F.E.F., L.B.; Project Administration: L.B.; Resources: A.S., K.L.W., L.B.; Software: W.T.G., M.v.A., L.E., A.H.; Visualization: W.T.G.; Writing – Original Draft: W.T.G.; Writing – Review & Editing: all coauthors

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Data and resource availability

Data related to the analyses in this study can be found at Data Dryad (DOI: XXXXX) or at https://github.com/wgough/XXXXXXXX. For more information, please contact the authors. All relevant data and details of resources can be found within the article and its supplementary information.

Tables

Table 1. Morphometrics for animals in each geographic location and demographic class.

Demographic Class	Sample Size		Body Length (cm)		Body Mass (kg)		Surface Area (m2)		Body Condition		Body Blubber Thickness (m)	
	Alaska	Hawaii	Alaska	Hawaii	Alaska	Hawaii	Alaska	Hawaii	Alaska	Hawaii	Alaska	Hawaii
Calf	126	1,033	7.96 ± 0.55	5.38 ± 0.64	8501 ± 2018	2475 ± 947	26.39 ± 3.99	11.70 ± 2.87	0.11571 ± 0.13879	0.00089 ± 0.11924	0.07483 ± 0.00383	0.05492 ± 0.00540
Juvenile	1	17	8.56 ± NA	8.39 ± 0.81	8409 ± NA	9388 ± 3312	27.41 ± NA	28.60 ± 6.31	-0.10000 ± NA	0.03235 ± 0.17711	0.07895 ± NA	0.07764 ± 0.00531
Yearling		21		9.01 ± 0.57		14329 ± 2856		36.58 ± 4.93		0.30905 ± 0.11528		0.08174 ± 0.00355
Lactating	142	1,013	12.97 ± 0.83	12.58 ± 0.78	29775 ± 5658	30719 ± 6404	63.44 ± 7.97	63.48 ± 8.37	-0.07310 ± 0.09150	0.04605 ± 0.12286	0.10127 ± 0.00301	0.09979 ± 0.00316
Adult	1,959	1,617	12.00 ± 0.94	11.37 ± 1.11	25771 ± 6411	21977 ± 6505	56.62 ± 9.15	51.01 ± 9.97	0.00009 ± 0.12156	-0.00966 ± 0.11855	0.09733 ± 0.00427	0.09436 ± 0.00556
Total	2,228	3,701	11.84 ± 1.34	10.00 ± 3.07	25041 ± 7476	18825 ± 12177	55.33 ± 11.48	43.27 ± 21.97	0.00192 ± 0.12531	0.01053 ± 0.12449	0.09630 ± 0.00679	0.08469 ± 0.01938

Figures

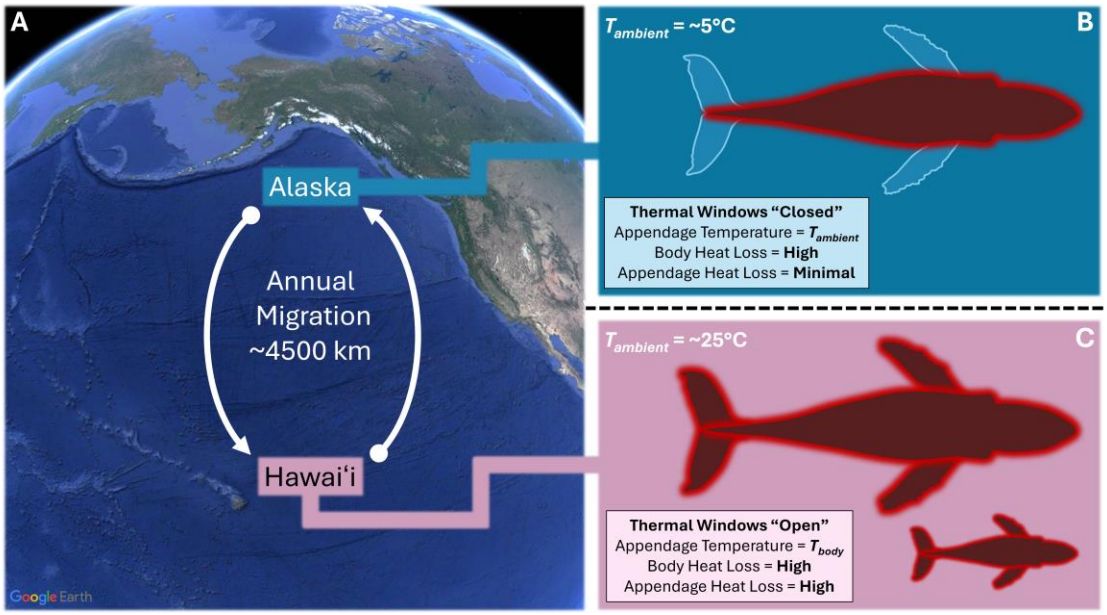


Figure 1: Schematic showing the relationship between geographic location and anatomical heat loss dynamics. A) Map of the North Pacific showing the migratory route for humpback whales between Alaska and Hawai'i. B) the heat loss dynamics for a hypothetical animal in Alaska with thermal windows (i.e., pectoral fins and tail flukes) closed. C) Similar heat loss dynamics for an animal in Hawai'i with thermal windows open and a calf present. Map image sourced from Google Earth version 7.3.6.10441.

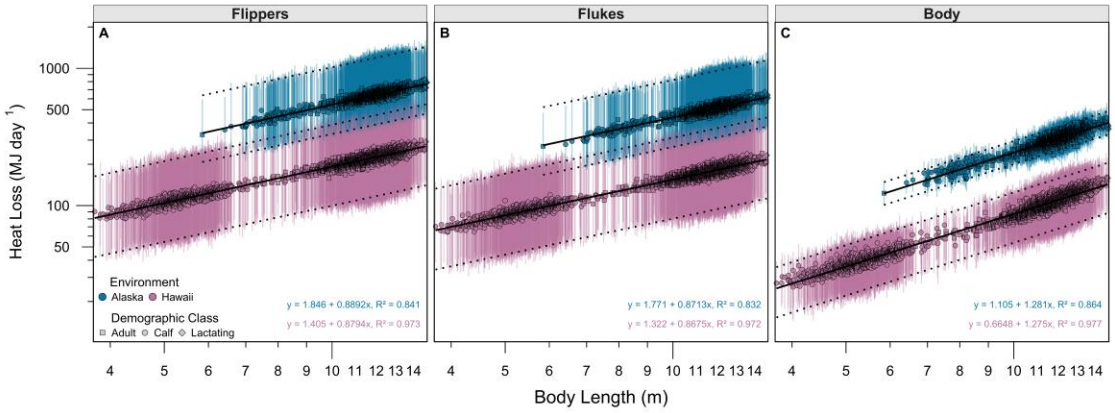


Figure 2: Relationships between body length and potential daily conductive heat loss from A) the flippers, B) the flukes, and C) the body. Data points for the flippers and flukes assume an open thermal state. Points denote median values for each individual, with vertical lines extending along the CI from 5% to 95%, based on our individual-scale Monte Carlo simulations. Regressions for all median and confidence interval values are denoted by solid black lines and dotted upper and lower boundaries, respectively.

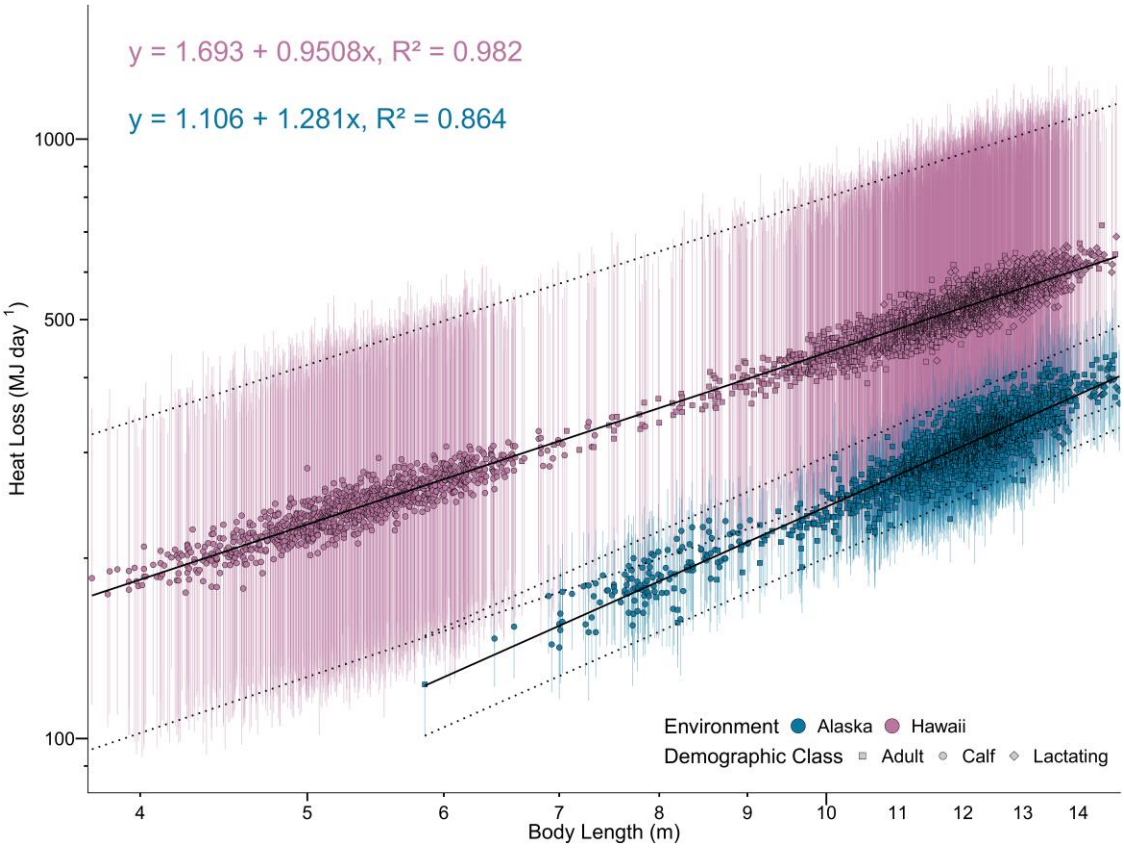


Figure 3: Relationship between body length and potential daily whole-animal heat loss for animals in Alaska (blue) with their thermal windows closed, and for animals in Hawai‘i (pink) with their thermal windows open. Points denote median values for each individual, with lines extending along the CI from 5% to 95%, based on our individual-scale Monte Carlo simulations. Dashed and dotted lines denote the regressions for those median and confidence interval values, respectively.

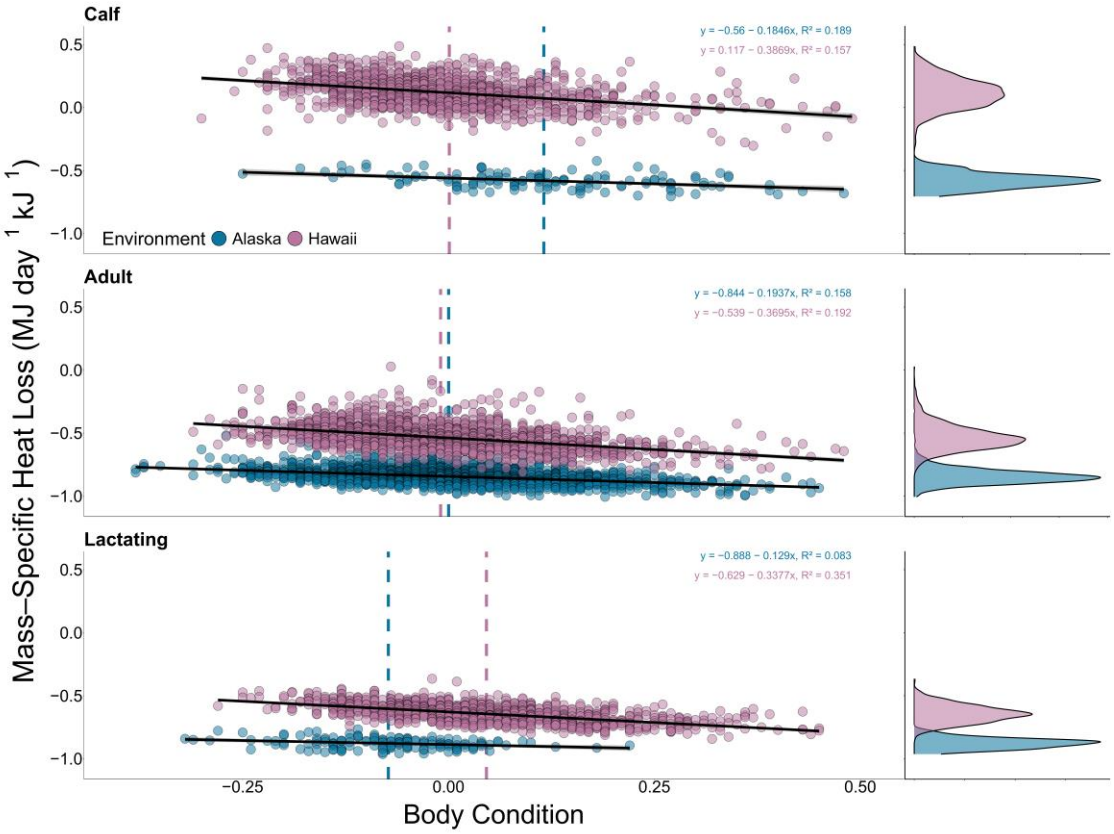


Figure 4: Relationship between body condition and mass-specific heat loss potential (MJ day^{-1} kJ^{-1}) with thermal windows open (Hawai'i; Pink) and closed (Alaska; Blue) for Calves (top), Adults (middle), and Lactating females (bottom). Vertical dashed lines denote the median value for each set of points.

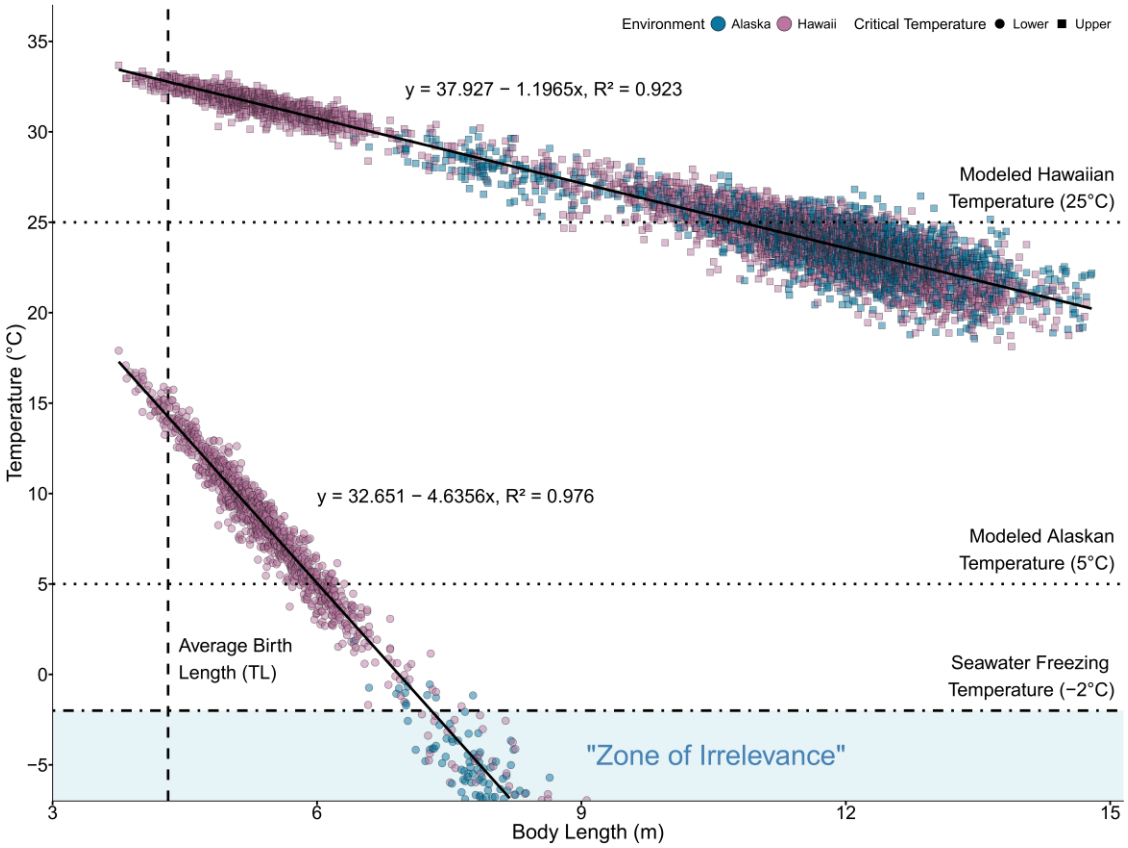


Figure 5: Relationship between body length and critical temperatures, assuming Kleiber BMR values for metabolic heat production. Horizontal dotted lines denote the temperatures (°C) used for our warm (Hawaiian) and cold (Alaskan) environments. The vertical dashed line denotes the mean estimated TL at birth (4.31 m) for calves in our dataset. The horizontal dot-dash line denotes the freezing temperature of seawater and the upper boundary of the "Zone of Irrelevance" described by Watts et al. (1993).

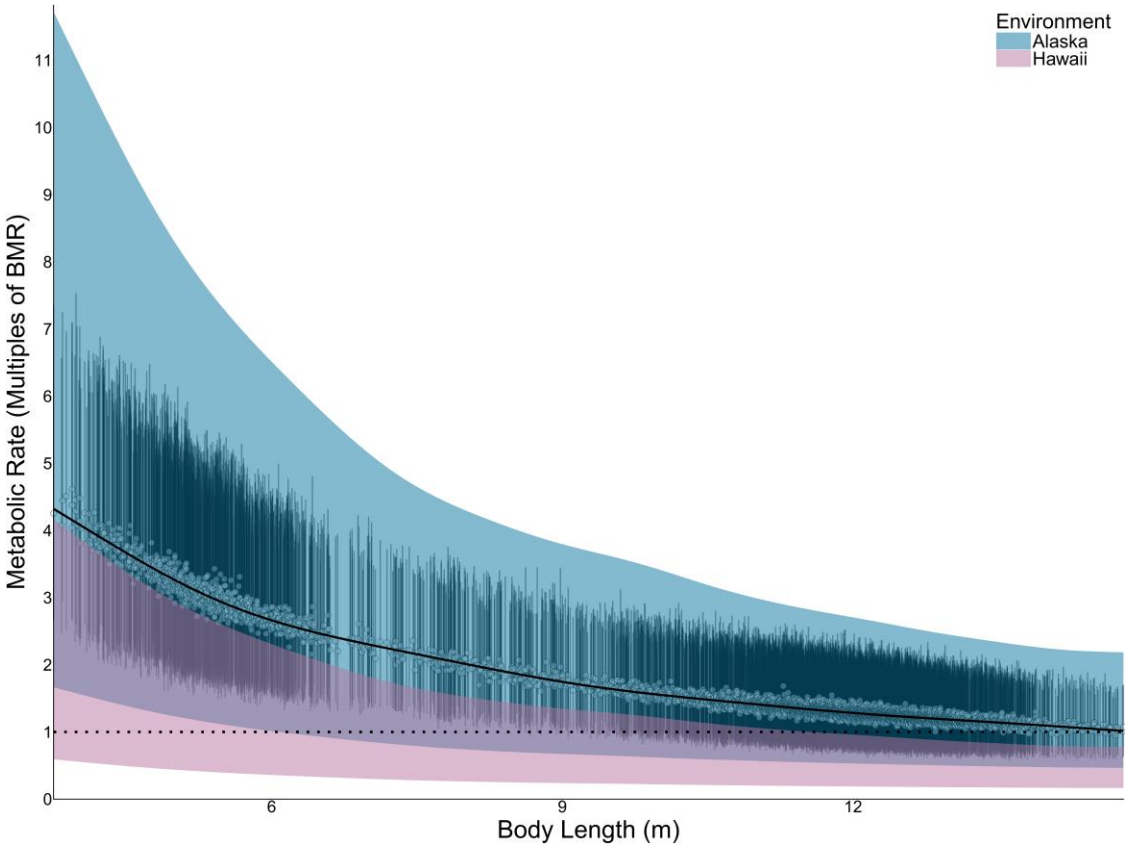


Figure 6: Relationship between body length and the range of potential heat loss values for both Alaska (blue) and Hawai'i (pink), displayed as multiples of "Kleiber" BMR (with the horizontal dashed line denoting 1x Kleiber BMR). The top and bottom of each range correspond to the potential heat loss with thermal windows open and closed, respectively. The darker purple portion is the overlap between the two ranges. Gray points denote median field metabolic rate estimates (derived from our breathing frequency regression) for each individual, with lines extending along the CI from 5% to 95%.

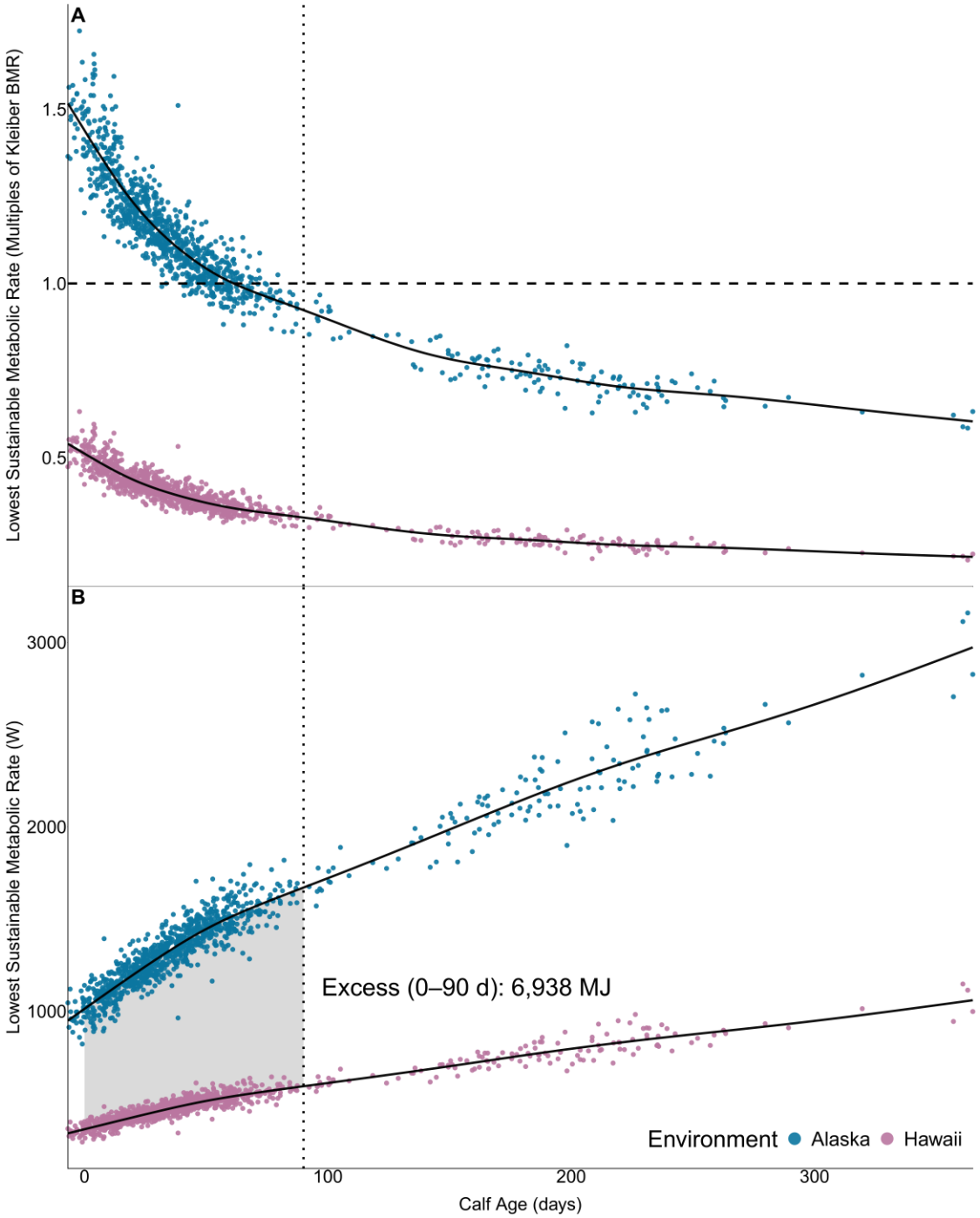


Figure 7: Age vs. the minimum sustainable metabolic rate for calves in Alaska (blue) and Hawai‘i (pink), displayed both as A) multiples of Kleiber BMR or B) absolute wattage. The horizontal dashed line denotes 1x Kleiber BMR, while the vertical dotted line corresponds to 90 days. The area of the shaded region has been calculated to show the cost differences for a calf growing in Hawai‘i vs. Alaska and maintaining its lowest sustainable metabolic rate.

Supplemental Materials

ID	Species	Sex	Age class	Stranding notes	Flipper / fluke	Body length (m)	Appendage blubber thickness (mm)
FA-6436	pygmy killer	Male	Subadult		Flipper	2.130	0.7000000
FA-9529	pygmy killer	Female	Subadult		Flipper	2.197	0.8000000
FA-9519	pygmy killer	Female	Adult	Pregnant	Flipper	2.290	1.7000000
FA-6599	pygmy killer	Female	Adult	Lactating	Flipper	2.320	3.5000000
FA-8456	pygmy killer	Male	Calf		Flipper	1.371	1.3000000
FA-2937	pygmy killer	Male	Adult	Emaciated	Flipper	2.320	0.8000000
FA-7995	pygmy killer	Male	Adult	Emaciated	Flipper	2.350	1.8000000
SC-3576	striped dolphin	Male	Adult		Flipper/Fluke	2.110	2.0500000
C-551	humpback whale	Unknown	Juvenile		Fluke	9.260	8.2741667
FEF-Plasticized	humpback whale	Male	Subadult		Flipper	9.020	3.0000000
HP2015001	harbor porpoise	Unknown	Unknown		Fluke	1.465	0.4695833

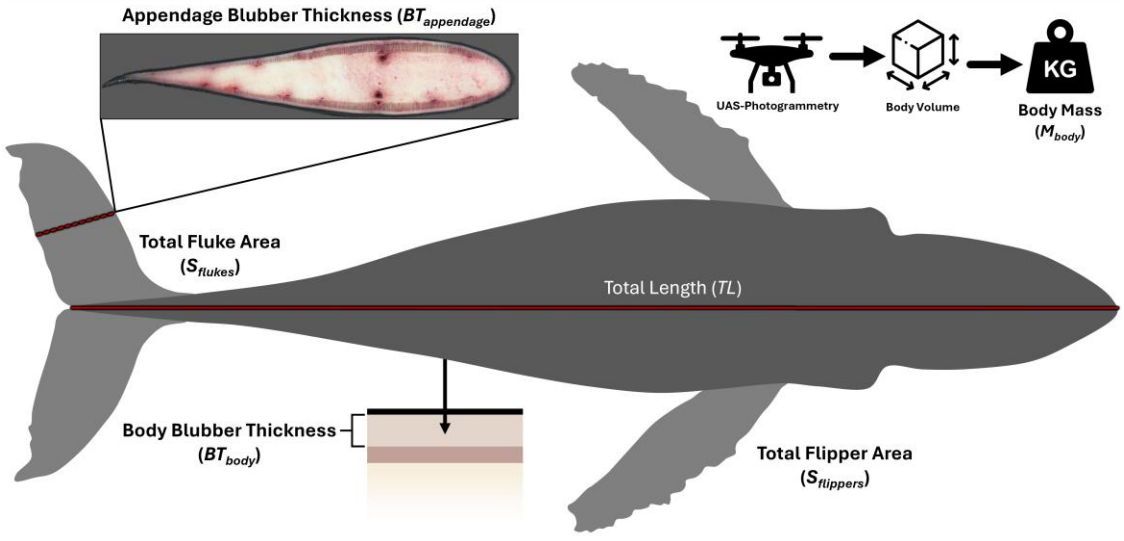


Figure S1: Schematic showing various morphometric and model-critical variables were measured. The cross-section (top left) is from the fluke of a harbor porpoise (C-489), showing the internal layers and vasculature, as well as the fusiform shape.

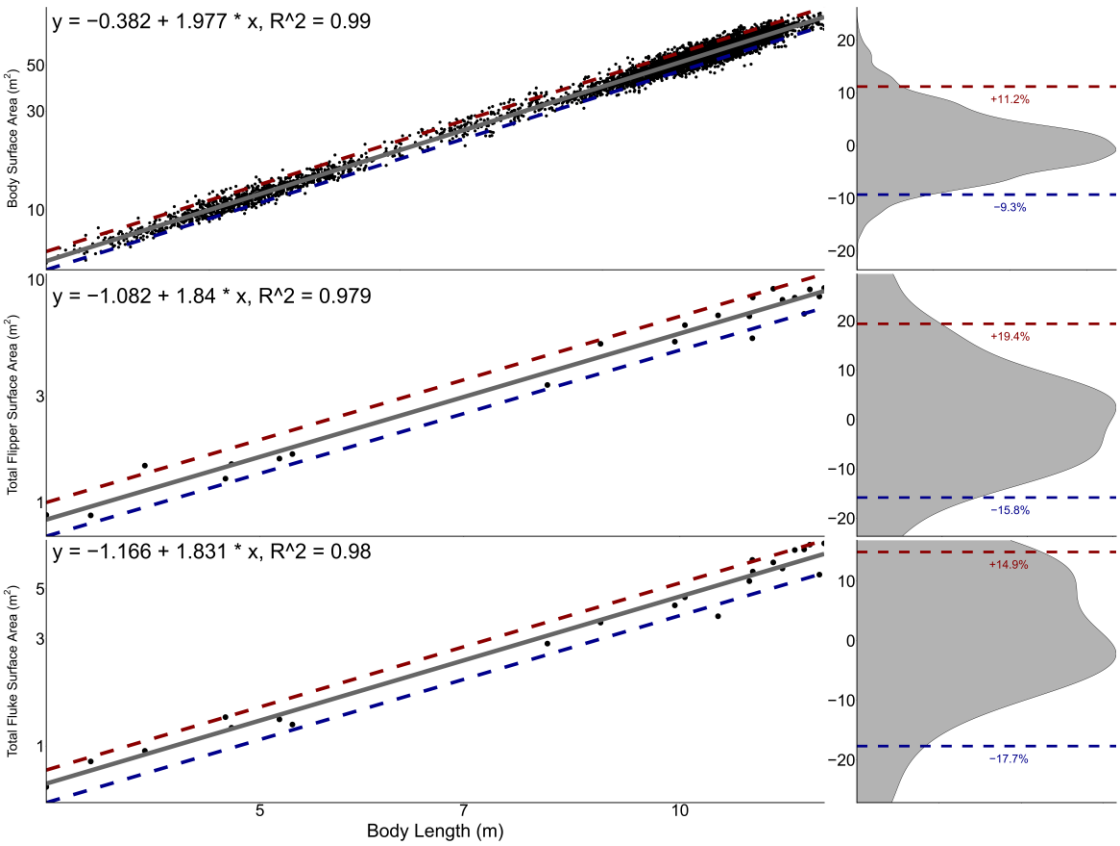


Figure S2: Morphometric relationships between body length and surface area estimates for A) the body, B) the flippers, and C) the flukes. Body values are taken from our broader dataset, while extremity values are taken from our subset of UAS data with clear flippers and flukes at the surface. Density plots show the residuals for each relationship (D-F), with the blue and red dotted lines denoting the 5% and 95% CI of each dataset, respectively.

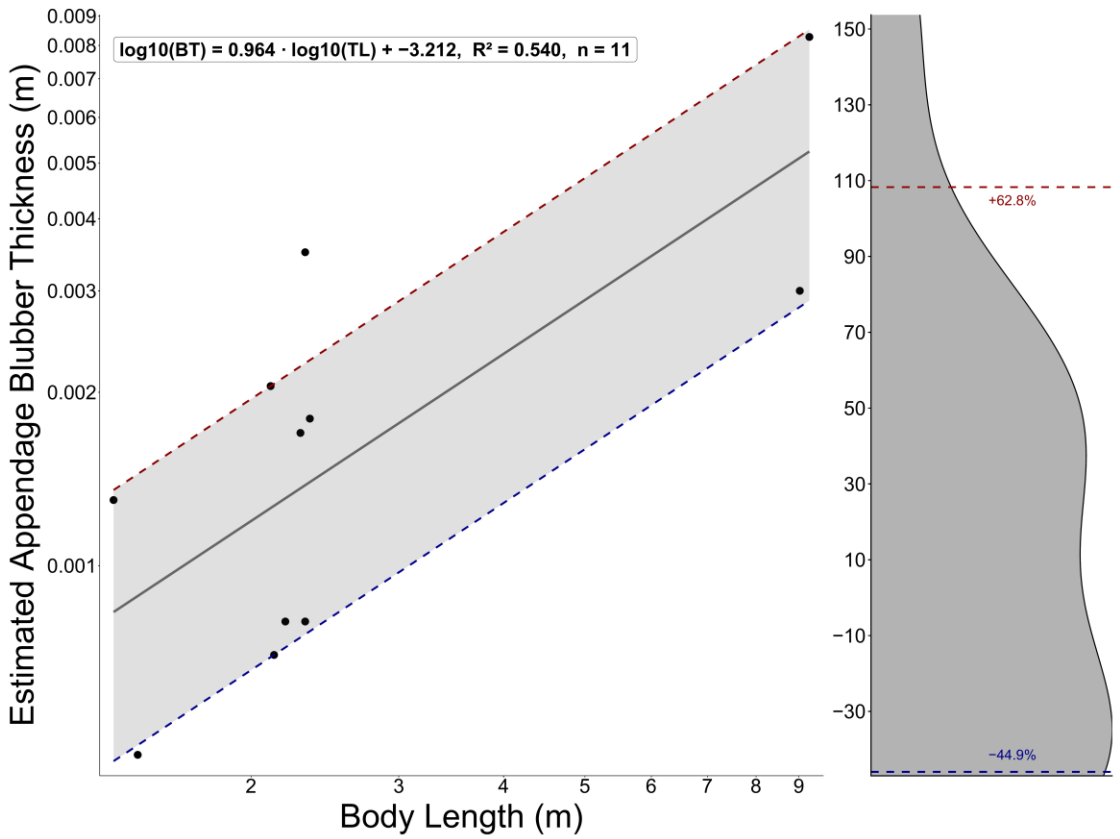


Figure S3: Morphometric relationships between body length and estimated appendage blubber thickness taken from our stranded animal dataset. Density plots show the residuals for each relationship (D-F), with the blue and red dotted lines denoting the 10% and 90% CI of each dataset, respectively.

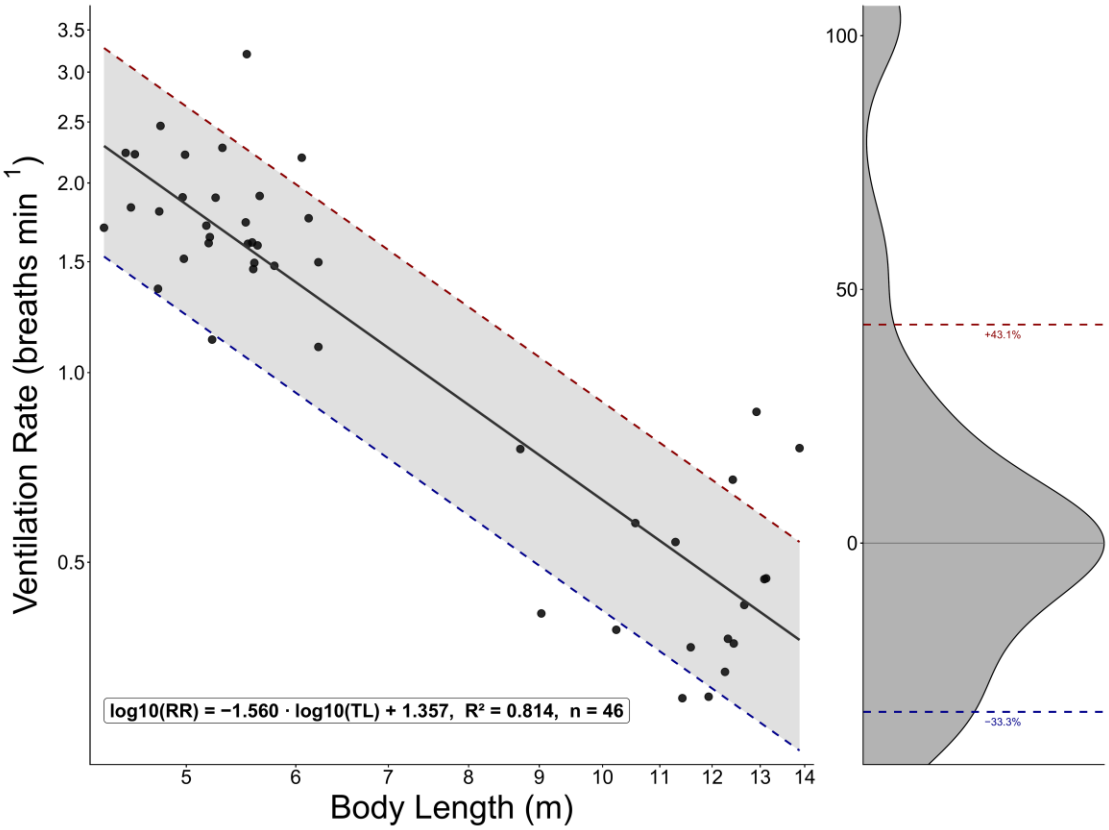


Figure S4: Log-transformed regression between body length and ventilation rate, taken from CATS tag data. The density plot shows the residuals for each individual, with the blue and red dotted lines denoting the 10% and 90% CI of each dataset, respectively.

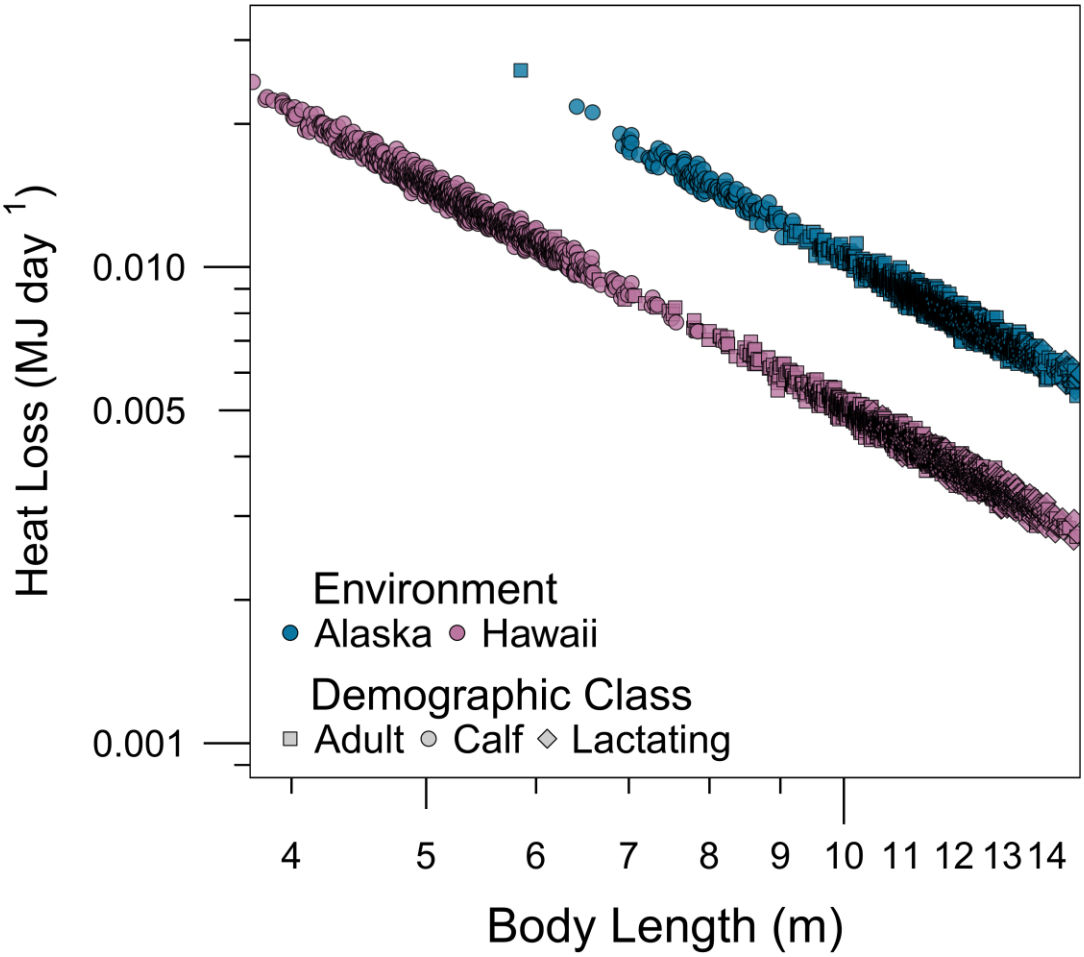


Figure S5: Ventilatory heat loss in Alaska (blue) and Hawai'i (pink).

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